INCH POUND

MIL-PRF-10304/19E
18 January 2008
SUPERSEDING
MIL-M-10304/19D
17 December 1975

PERFORMANCE SPECIFICATION SHEET

METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED:
SPECIAL DECIBEL (FLUSH MOUNTING, SQUARE FLANGE, 1.5 INCH) STYLE 13

Inactive for new design after November 4, 1997

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-10304.

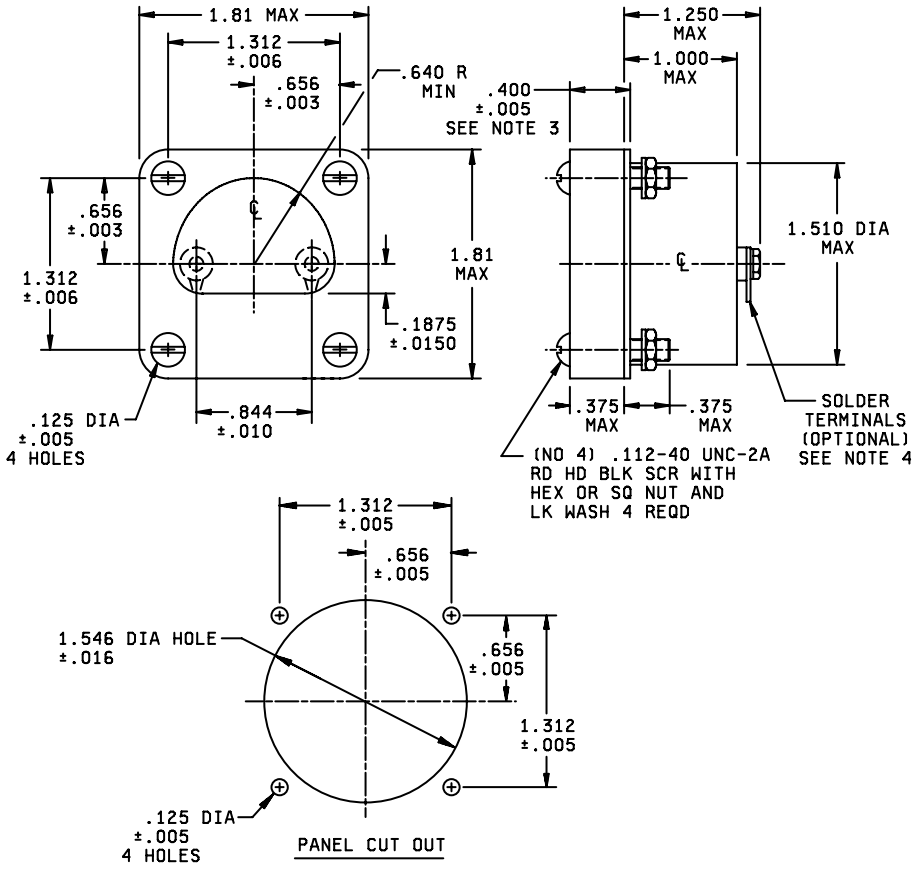



FIGURE 1. Decibel meter, 1.5 inch.

AMSC N/A

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .003 | .08 | .016 | .41 | .640 | 16.26 | 1.312 | 33.32 |
| .005 | .13 | .125 | 3.18 | .656 | 16.70 | 1.510 | 38.35 |
| .006 | .15 | .1875 | 4.76 | .844 | 21.44 | 1.546 | 39.28 |
| .010 | .25 | .375 | 9.53 | 1.000 | 25.40 | 1.81 | 45.97 |
| .015 | .38 | .400 | 10.16 | 1.250 | 31.75 | | |

NOTES:
1. Dimensions are in inches. Unless otherwise specified, the tolerances are ± .005 (.13 mm).
2. Metric equivalents are given for general information only.
3. This dimension shall be the maximum projection from front of panel, including mounting gaskets (if used).
4. Solder terminals shall be used.
5. The centerline of the terminals shall be equally spaced on each side of the vertical centerline and located within .188 (4.78 mm) above or below the centerline of the meter.
6. Other forms of terminals, as specified in MIL-PRF-10304, are permitted. The specified minimum length applies to stud-type terminals only. Terminals shall be equally spaced, from each side of vertical centerline.

FIGURE 1. Decibel meter, 1.5 inch – Continued.

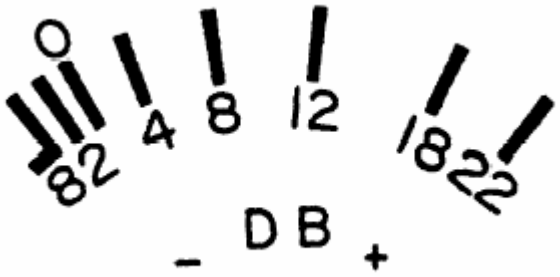



FIGURE 2. Scale plate design.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 4 ounces maximum.

Full-scale ranges: See table I.

Scale:
Length: 1 inch minimum.
Pointer deflection: 90 degrees nominal.

Color scheme: B

Pointer: Lance type, white non-radioactive.

Range: - 8 to + 22 dB (see figure 2). Reference level 1 milliwatt in 600 ohms.

Voltage: 0.775 volt, 1,000 Hz at 0-dB scale position.

Actual level: 0 at 0-dB scale position from zero power level.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

Position influence: Not greater than ± 2 percent of full scale length for 60 degrees rotation from normal operating position.

Initial Accuracy: Not to exceed the values shown in table I.

Overshoot: 10 percent maximum.

Response time: 0.5 second maximum at 1,000 Hz at the 0-dB scale position.

Effective resistance: 5,000 ohms minimum.

Frequency range: 25 to 10,000 Hz (the change in indication at scale points - 8, 0, + 8, + 12, and + 22 dB shall not exceed the values shown in table II).

Special meter: Serial number 100.

Heat effect: At + 65 degrees C, to be not greater than shown in table III.

Temperature influence: Not greater than shown in table IV.

Exposure to extreme temperatures: Change in indication not to exceed the values shown in table V.

Sustained overload: Not to exceed the values shown in table VI.

Vibration:
Random drop: } Permanent change in indication after each test shall not exceed the values shown in table VII.
Shock:

Dielectric withstanding voltage: 1,500 volts rms, without damage or flashover.

TABLE I. Initial accuracies.

| Scale position 1/ | Equivalent voltage with 1,000 Hz sine wave 1/ | Error tolerance | Min-max point | Equivalent voltage 1/ |
|---|---|---|---|---|
| dB | Volts | dB | dB | Volts |
| - 8 | 0.308 | ± 7 | - 15 to - 1 | 0.137 to 0.690 |
| - 2 | 0.615 | ± 5 | - 7 to + 3 | 0.346 to 1.094 |
| 0 | 0.775 | ± 1 | - 1 to + 1 | 0.690 to 0.869 |
| + 4 | 1.228 | ± 2 | + 2 to + 6 | 0.975 to 1.542 |
| + 8 | 1.947 | ± 2 | + 6 to + 10 | 1.543 to 2.449 |
| + 12 | 3.084 | ± 2 | + 10 to + 14 | 2.449 to 3.885 |
| + 18 | 6.153 | ± 0.75 | + 17.25 to + 18.75 | 5.653 to 6.700 |
| + 22 | 9.751 | ± 0.50 | + 21.50 to + 22.50 | 9.210 to 10.320 |

1/ The above values are based upon the expression: $dB = 20 \log_{10}\{\text{Equiv. Voltage} / .775\}$.

TABLE II. Frequency range.

| From (Hz) | To (Hz) | Change in indication (dB) |
|---|---|---|
| 25 | 1,000 | ± 0.5 |
| 1,000 | 5,000 | ± 1.0 |
| 5,000 | 10,000 | ± 3.5 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

TABLE III. Heat effect.

| Scale position (dB) | Temporary (dB) | Permanent change in indication (dB) |
|---|---|---|
| - 8 | ± 5.0 | ± 5.0 |
| 0 | ± 2.0 | ± 1.0 |
| + 8 | ± 1.5 | ± 1.0 |
| + 12 | ± 1.5 | ± 1.0 |
| + 22 | ± 1.0 | ± 0.5 |

TABLE IV. Temperature influence.

| Scale position (dB) | Change in indication (dB) |
|---|---|
| - 8 | ± 3 |
| 0 | ± 2 |
| + 8 | ± 2 |
| + 12 | ± 2 |
| + 22 | ± 1 |

TABLE V. Exposure to extreme temperature.

| Scale position (dB) | Change in indication (Limited to + 65° C, and - 55° C) | Permanent change in indication (dB) |
|---|---|---|
| - 8 | Free indication 1/ | ± 4 |
| 0 | Free indication | ± 2 |
| + 8 | Free indication | ± 2 |
| + 12 | Free indication | ± 2 |
| + 22 | Free indication | ± 1 |

1/ See MIL-PRF-10304, section 6, definition of “Free indication”.

TABLE VI. Sustained overload.

| Scale position (dB) | Permanent change in indication (dB) |
|---|---|
| - 8 | ± 4.0 |
| 0 | ± 1.5 |
| + 8 | ± 1.5 |
| + 12 | ± 1.5 |
| + 22 | ± 1.5 |

TABLE VII. Vibration, random drop, and shock.

| Scale position (dB) | Permanent change in indication (dB) |
|---|---|
| - 8 | ± 6.0 |
| 0 | ± 3.5 |
| + 8 | ± 2.0 |
| + 12 | ± 2.0 |
| + 22 | ± 2.0 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.

The Qualified Products List (QPL) associated with this inactive for new design specification sheet will be maintained until acquisition of the product is no longer required, whereupon the specification and QPL will be canceled.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. This document references MIL-PRF-10304.

## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 6625-0941-000)

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - CG, MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at **http://assist.daps.dla.mil**.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:50Z
Check the source to verify that this is the current version before use.